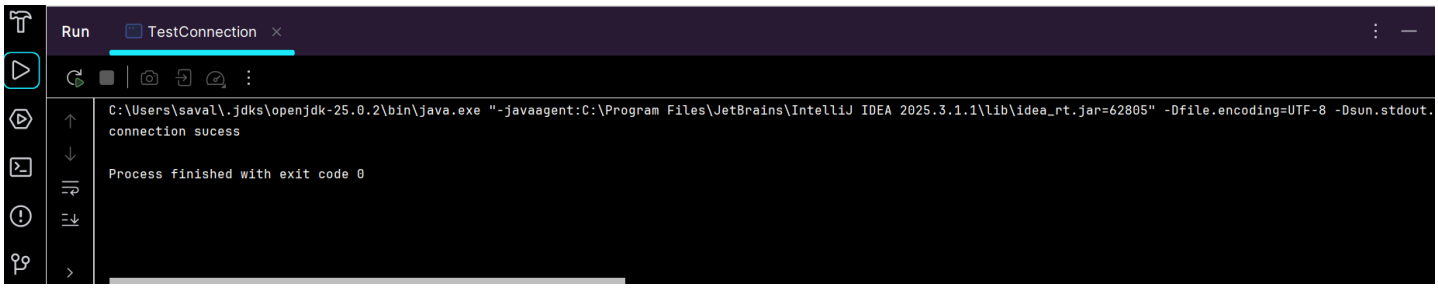
Code :-

Testconnection:-

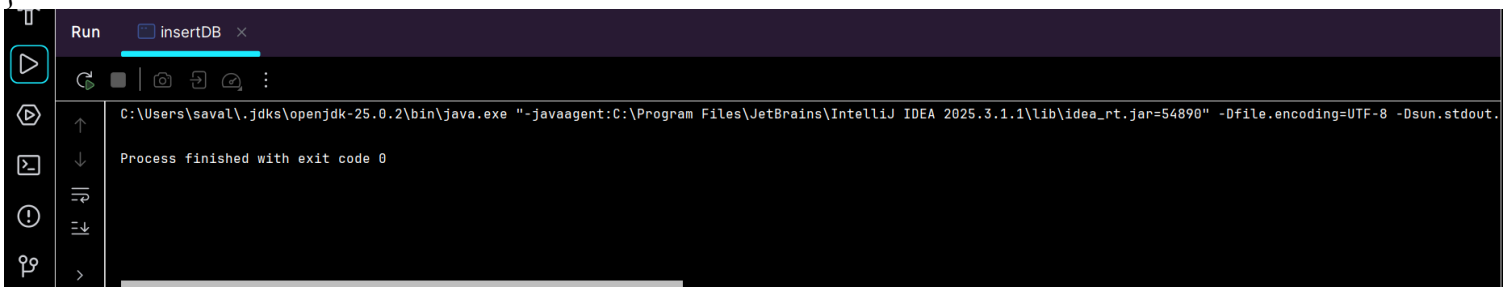
```
//Write JDBC program to test connection
import java.sql.DriverManager;//Loads
import java.sql.Connection;// Connection

public class TestConnection
{
   public static void main(String[] args) throws Exception
   {
      Class.forName("com.mysql.cj.jdbc.Driver");//Register the driver class
      String url="jdbc:mysql://localhost:3306/";//create connection
      String user="root";
      String pwd="root";
//       Connection con=DriverManager.getConnection(url,user,pwd);
      Connection con=DriverManager.getConnection(url,"root","");
      if(con!=null)
         System.out.println("connection sucess");
      else
         System.out.println("fail");

      con.close();
   }
}
```



Insert Database:-

```
//Create a table
import java.sql.DriverManager;//Loads
import java.sql.Connection;// Connection
import java.sql.Statement;
public class insertDB
{
   public static void main(String[] args) throws Exception {
      Class.forName("com.mysql.cj.jdbc.Driver");//1. Register the driver class
      String url = "jdbc:mysql://localhost:3306/productdb";//2. create connection
      String pwd = "root";
      Connection con = DriverManager.getConnection(url, "root", "");
```

```
        String query = "create table product(ID int," + "pname varchar(30),price int)";//3. Create query
        Statement st = con.createStatement();
        if (st.execute(query))//4. Execute Query
            System.out.println("Table Created");
        con.close();//colse the quary
        st.close();
    }
}
```



InsertUpdates:-

```
import java.io.IOException;
import java.sql.*;

public class insertUpdate
{
    public static void main(String[] args) throws Exception
    {
        Class.forName("com.mysql.cj.jdbc.Driver");//1. Register the driver class
        String url = "jdbc:mysql://localhost:3306/productdb";//2. create connection
        String pwd = "root";
        Connection con = DriverManager.getConnection(url, "root", "");
        String Query="insert into product values(101,'mobile',10000)";
        Statement st=con.createStatement();
            int i=st.executeUpdate(Query);
            if(i>0) {
                System.out.println("record inserted");
            }
            else {
                System.out.println("Check sql syntex");
            }
            con.close();
            st.close();
    }

}
```

```
Run    insertUpdate
C:\Users\saval\.jdks\openjdk-25.0.2\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.1.1\lib\idea_rt.jar=55990" -Dfile.encoding=UTF-8 -Dsun.stdout.
record inserted

Process finished with exit code 0
```

Fetch:-

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;
import java.sql.ResultSet;

public class fetch
{
   public static void main(String[] args) throws Exception {
      Class.forName("com.mysql.cj.jdbc.Driver");//1. Register the driver class
      String url = "jdbc:mysql://localhost:3306/productdb";//2. create connection
      String pwd = "root";
      Connection con = DriverManager.getConnection(url, "root", "");
      String Query="select * from product";
      Statement st=con.createStatement();
         ResultSet rs=st.executeQuery(Query);
         System.out.println("SrNo ID   pname   price");
         int r=1;
         while(rs.next()){
            System.out.print(r+" ");
            System.out.print(rs.getInt("ID")+" ");
            System.out.print(rs.getString("pname")+" ");
            System.out.println(" ");
            r++;
         }
         rs.close();
         con.close();
         st.close();
   }
}
```

```
Run    fetch
C:\Users\saval\.jdks\openjdk-25.0.2\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.1.1\lib\idea_rt.jar=59256" -Dfile.encoding=UTF-8 -Dsun.stdout.
ID pname price
101 0mobile  102 1mobile
Process finished with exit code 0
```

Conclusion:-

In this practical, we successfully implemented JDBC (Java Database Connectivity) to interact with a MySQL database. We learned how to register the JDBC driver, establish a database connection, create a table, insert records, and retrieve data from the database using Java programs. The output was correctly displayed on the console, which confirms proper communication between the Java application and the database. This practical helps in understanding the basic concepts of database connectivity in Advanced Java and builds a strong foundation for developing database-driven applications.